

GEARS Human Validation — Consensus & Coordinator

Purpose

Preserve both independent human records and provide one place to establish the consensus human reference standard for the 50 GEARS validation studies.

Source records

Reviewer 1 workspace: reviewer_1_blind_form_simple_GEARs

Reviewer 2 workspace: reviewer_2_blind_form_simple_GEARs

Both source documents remain unchanged.

Current comparison

Record identity: 50 of 50 study IDs are present in both reviewer documents.

Normalized core-field match before adjudication: 41 of 50 records.

Human adjudication completed: 8 records (GEARS-HV-005, GEARS-HV-017, GEARS-HV-018, GEARS-HV-035, GEARS-HV-040, GEARS-HV-042, GEARS-HV-043, GEARS-HV-047).

Substantive reconciliation remaining: 1 record (GEARS-HV-020).

Terminology cleanup only: GEARS-HV-015 and GEARS-HV-038 use “Extend” in Reviewer 1 and “Extend or develop” in Reviewer 2.

Lock note: Reviewer 2’s Review status field is blank for all 50 records; Reviewer 1 has 49 LOCKED records and one blank status (GEARS-HV-047).

Confirm the independent-review lock before freezing consensus.

Vocabulary note: Reviewer 2 often entered “Yes” for evidence sufficiency; this form treats it as equivalent to “Sufficient” for comparison but preserves the original wording below.

Workflow

Confirm both independent records are locked.

For rows marked MATCH, verify the prefilled records and enter the consensus result.

Rows marked RESOLVED contain the entered human adjudication and need no further substantive review.
For rows marked RECONCILE, discuss only the flagged differences or missing values.
Use REFERENCE_UNRESOLVED if agreement cannot be reached; never force a label.
Freeze the consensus before linking back to machine labels or the private sampling crosswalk.

Allowed values

Consensus status: AGREED / RECONCILED / REFERENCE_UNRESOLVED
Citation validity: Yes / No / Unclear
Method executed: Yes / No / Unclear
Uses (select all): Mention only / Benchmark / Extend or develop / Apply to biological data / Other executed use / Unclear
Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear
Biological insight: Yes / No / Unclear
Evidence sufficiency: Sufficient / Incomplete / Citation not located / Unclear

Consensus table

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-001 FORMAL VALIDATION Modeling interpretable correspondence between cell state and perturbation response with CellCap DOI: 10.1101/2024.03.14.585078 Article page: https://doi.org/10.1101/2024.03.14.585078 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/16/2024.03.14.585078.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: or example, scGen uses a deep generative model to predict 67 the impact of one perturbation on a new cell population [10], and GEARS takes advantage of	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For example, scGen uses a deep generative model to predict the impact of one perturbation on a new cell population [10], and GEARS takes advantage of prior biological knowledge to model nonlinear gene	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	68 prior biological knowledge to model nonlinear gene interactions and nonlinearities in response 69 to multiple perturbations [11]. Notes:	interactions and nonlinearities in response to multiple perturbations [11] Notes:	
GEARS-HV-002 FORMAL VALIDATION Single-cell technologies: current and near future DOI: 10.1007/s11427-024-2813-y Article page: https://doi.org/10.1007/s11427-024-2813-y Full text 1: https://link.springer.com/content/pdf/10.1007/s11427-024-2813-y.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For scCRISPR data analysis, existing bioinformatics tools generally focus on three aspects, including data preprocessing (Wei et al., 2023), data denoising (Duan et al., 2019), and functional analysis (Roohani et al., 2024), aiming to uncover rich perturbation information. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: One of the most impressive applications of single-cell multi-omic technologies is the joint profiling of mRNA expression and epigenomic states. For example Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-003 FORMAL VALIDATION rbio1 - training scientific reasoning LLMs with biological world models as soft verifiers DOI: 10.1101/2025.08.18.670981 Article page: https://doi.org/10.1101/2025.08.18.670981 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/21/2025.08.18.670981.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compare against: SUMMER [32] (retrieval + knowledge-based reasoning), GEARS [38] (a specialized perturbation model), and the base Qwen2.5-3B reasoning model without biological supervision. Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare against: SUMMER [32] (retrieval + knowledge-based reasoning), GEARS [38] (a specialized perturbation model), and the base Qwen2.5-3B reasoning model without biological supervision. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-004 FORMAL VALIDATION Multimodal Benchmarking of Foundation Model Representations for Cellular Perturbation Response Prediction DOI: 10.1101/2025.06.26.661186 Article page: https://doi.org/10.1101/2025.06.26.661186 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/06/28/2025.06.26.661186.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: There are many specialized perturbation response models one could use on top of such embeddings (e.g. Roohani et al. (2024), Lotfollahi et al. (2023), Piran et al. (2024)), but those models are beyond the scope of our work, which intentionally fixes a single model with a consistent training protocol. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: There are many specialized perturbation response models one could use on top of such embeddings (e.g. Roohani et al. (2024), Lotfollahi et al. (2023), Piran et al. (2024)), but those models are beyond the scope of our work, which intentionally fixes a single model with a consistent training protocol. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-005 FORMAL VALIDATION GexMolGen: cross-modal generation of hit-like molecules via large language model encoding of gene expression signatures DOI: 10.1093/bib/bbae525 Article page: https://doi.org/10.1093/bib/bbae525 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/11/26/2023.11.11.566725.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Instead of commonly using mean square loss, we use the autofocus direction-aware loss in GEARS [32]. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We further utilize the mask language model (MLM) task to fine-tune scGPT. Instead of commonly using mean square loss, we use the autofocus direction-aware loss in GEARS [32]. Notes:	Comparison flag: RESOLVED — human adjudication selected Mention only. Consensus status: RECONCILED Citation validity: Yes Method executed: No Uses (select all): Mention only Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We further utilize the mask language model (MLM) task to fine-tune scGPT. Instead of commonly using mean square loss, we use the autofocus direction-aware loss in GEARS [32]. Resolution note: Human adjudication determined that GEARS itself was not executed; the paper only referred to a loss associated with GEARS.
GEARS-HV-006 FORMAL VALIDATION	Review status: LOCKED Citation validity: Yes	Review status: Citation validity: Yes	Comparison flag: MATCH — normalized core fields agree.

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Disentanglement of single-cell data with biolord DOI: 10.1038/s41587-023-02079-x Article page: https://doi.org/10.1038/s41587-023-02079-x Full text 1: https://www.nature.com/articles/s41587-023-02079-x.pdf	Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We show that biolord outperforms GEARS in the prediction of unseen one-gene perturbation (normalized mean squared error, one of one gene unseen; GEARS: 0.47; biolord: 0.37) and two-gene perturbations (normalized mean squared error, two of two genes unseen; GEARS: 0.53; biolord: 0.50, one of one gene unseen; GEARS: 0.39; biolord: 0.35, zero of two genes unseen; GEARS: 0.28; biolord: 0.20; Methods; Supplementary Note 3 and Extended Data Fig. 2). Notes:	Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We show that biolord outperforms GEARS in the prediction of unseen one-gene perturbation (normalized mean squared error, one of one gene unseen; GEARS: 0.47; biolord: 0.37) and two-gene perturbations (normalized mean squared error, two of two genes unseen; GEARS: 0.53; biolord: 0.50, one of one gene unseen; GEARS: 0.39; biolord: 0.35, zero of two genes unseen; GEARS: 0.28; biolord: 0.20; Methods; Supplementary Note 3 and Extended Data Fig. 2). Notes:	Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-007 FORMAL VALIDATION Impact of the Human Cell Atlas on medicine DOI: 10.1038/s41591-022-02104-7 Article page: https://doi.org/10.1038/s41591-022-02104-7 Full text 1: https://www.nature.com/articles/s41591-022-02104-7.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Moreover, machine-learning algorithms can increasingly be trained on such data to yield models that predict the impact of additional perturbations in one or more genes in the same cellular context or of the same perturbations in new biological contexts^{114,115,116}. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Moreover, machine-learning algorithms can increasingly be trained on such data to yield models that predict the impact of additional perturbations in one or more genes in the same cellular context or of the same perturbations in new biological contexts^{114,115,116}. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-008 FORMAL VALIDATION scFormer: A Universal Representation Learning	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all):	Review status: Citation validity: Yes Method executed: Yes Uses (select all):	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Approach for Single-Cell Data Using Transformers DOI: 10.1101/2022.11.20.517285 Article page: https://doi.org/10.1101/2022.11.20.517285 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2022/11/22/2022.11.20.517285.full.pdf	Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compare the performance against the recent GEARS method (Roohani et al., 2022) and the multi layer perceptron baseline. Notes:	Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare the performance against the recent GEARS method (Roohani et al., 2022) and the multi layer perceptron baseline. Notes:	Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-009 FORMAL VALIDATION Unagi: Deep Generative Model for Deciphering Cellular Dynamics and In-Silico Drug Discovery in Complex Diseases DOI: 10.21203/rs.3.rs-3676579/v1 Article page: https://doi.org/10.21203/rs.3.rs-3676579/v1 Full text 1: https://www.researchsquare.com/article/rs-3676579/latest.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Although GEARS25 and scGen23 are able to perform in-silico perturbations, they weren't designed to process time-series data and require the in-vitro screening of perturbation response of disease and drug treatments as the supervision. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: nt gap in current single-cell methodologies is the absence of 74 unsupervised in-silico exploration capabilities. Although GEARS 25 and scGen 23 are able to perform in-silico 75 perturbations, they weren't designed to process time-series data and require the in-vitro screening of 76	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
		perturbation response of disease and drug treatments as the supervision. Thus, there is a pressing need for 77 methods that can virtually examine thousands of potential drugs and compounds on single-cell disease data 78 without ground truth training data. Notes:	
GEARS-HV-010 FORMAL VALIDATION On the use of variational autoencoders for biomedical data integration DOI: 10.1101/2025.08.18.670835 Article page: https://doi.org/10.1101/2025.08.18.670835 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/22/2025.08.18.670835.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: These values were comparable to those of the other models benchmarked in a single gene KD setting by Ahlmann-Eltze et al., including foundation models such as scGPT (L2RPE1=6.88, L2K562=4.41) graph-based models like GEARS (L2RPE1=6.33, L2K562=4.04), and their proposed bilinear baseline (L2RPE1=5.84, L2K562=4.04) (Figure 6a,b, Supplementary Figure 16). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: we compared MOVE's performance with that of alternative methods including GEARS, CPA, scGPT and their baseline linear models. T Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-011 FORMAL VALIDATION	Review status: LOCKED Citation validity: Yes Method executed: No	Review status: Citation validity: Yes Method executed: No	Comparison flag: MATCH — normalized core fields agree. Consensus status:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
scEMB: Learning context representation of genes based on large-scale single-cell transcriptomics DOI: 10.1101/2024.09.24.614685 Article page: https://doi.org/10.1101/2024.09.24.614685 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/09/26/2024.09.24.614685.full.pdf	Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Moreover, these embedding-based methods¹¹ [Autoencoder¹², VAE^{13–16}, GNN¹⁷], has been widely used to along with the cellular perturbation tasks along with single-cell CRISPR data, to compress the transcriptomics data (mainly single-cell) into the latent space and compare the effects of different perturbations and possess a more accurate molecular measurement as well as in silico-induced perturbations^{18,19}. Notes:	Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: ecent advances in artificial intelligence (AI), such as variational autoencoder (VAE) models^{8–10}, which explicitly learn low-dimensional embeddings from single-cell transcriptome to capture the biological meaning embedding. Moreover, these embedding-based methods¹¹ [Autoencoder¹², VAE^{13–16}, GNN¹⁷], Notes:	Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-012 FORMAL VALIDATION Evaluating the Utilities of Large Language Models in Single-cell Data Analysis DOI: 10.21203/rs.3.rs-3376641/v1 Article page: https://doi.org/10.21203/rs.3.rs-3376641/v1 Full text 1: https://www.researchsquare.com/article/rs-3376641/latest.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We considered scGPT and GEARS [42] for this task. In the training process, we masked the genes under perturbation and tried to reconstruct the expression levels of all the genes of the input cells rather than only the masked genes. Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We considered scGPT and GEARS [42] for this task. In the training process, we masked the genes under perturbation and tried to reconstruct the expression levels of all the genes of the input cells rather than only the	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
		masked genes. We used Mean Pearson Correlation (MPC) as the metric to evaluate the performance of scGPT under different hyper-parameters or initial settings, and details can be found in Appendix C . T Notes:	
GEARS-HV-013 FORMAL VALIDATION A systematic comparison of computational methods for expression forecasting DOI: 10.1101/2023.07.28.551039 Article page: https://doi.org/10.1101/2023.07.28.551039 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/07/31/2023.07.28.551039.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We included all methods with peer-reviewed paper and code available as of Sept 1, 2023, which were DCD-FG, GEARS, and GeneFormer (Lopez et al., 2022; Roohani et al., 2022; Theodoris et al., 2023). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For adamson, dixit, and norman, we eschew these filters and instead use preprocessing identical to the GEARS evaluations (Roohani et al., 2024). Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-014 FORMAL VALIDATION scMomer: A modality-aware pretraining framework for single-cell multi-omics modeling under missing modality conditions DOI: 10.1101/2025.08.04.668374	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Following the setup used in CellFM and scFoundation, we integrated gene	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Results showed that scMomer outperformed GEARS, achieving improvements of	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Article page: https://doi.org/10.1101/2025.08.04.668374 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/05/2025.08.04.668374.full.pdf	embeddings from scMomer into the GEARs13 for downstream prediction (Supplementary Fig. 11). Notes:	0.034 in PCC (Supplementary Table 5). Notes:	Resolution note:
GEARS-HV-015 FORMAL VALIDATION MultiPert: An adversarial alignment and dual attention framework for single-cell multi-omics perturbation prediction DOI: 10.1371/journal.pcbi.1014054 Article page: https://doi.org/10.1371/journal.pcbi.1014054 Full text 1: https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1014054&type=printable	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Given GEARs' prior validation of GO knowledge graph for perturbation prediction, we directly adopted their pre-trained embeddings to initialize MultiPert's perturbation representations [16]. Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: . Given GEARs' prior validation of GO knowledge graph for perturbation prediction, we directly adopted their pre-trained embeddings to initialize MultiPert's perturbation representations [16]. Notes:	Comparison flag: MATCH — terminology normalization needed ("Extend" → "Extend or develop"). Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-016 FORMAL VALIDATION Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation DOI: 10.1038/s41587-025-02777-8 Article page: https://doi.org/10.1038/s41587-025-02777-8 Full text 1: https://www.nature.com/articles/s41587-025-02777-8.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We considered three state-of-the-art perturbation response prediction methods: compositional perturbation autoencoder (CPA)10, GEARs1 and scGPT2 (Methods). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We next evaluated performance on the task of predicting transcriptional responses to two-gene perturbations in the Norman14 dataset, the only dataset with combinatorial perturbations in our benchmark. For unseen two-gene perturbations (two-gene perturbations where none of the individual perturbations was observed at train	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
		time), the matching mean baseline outperformed all other baselines by a considerable margin, with relative improvements of 11% for Pearson Δ over the best alternative method (GEARS; Fig. 1d). Notes:	
GEARS-HV-017 FORMAL VALIDATION A General Single-Cell Analysis Framework via Conditional Diffusion Generative Models DOI: 10.1101/2023.10.13.562243 Article page: https://doi.org/10.1101/2023.10.13.562243 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/17/2023.10.13.562243.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend/Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Under such zero-shot setting, we incorporate biological prior for the unseen genes by adapting the gene ontology-based graph neural network from GEARS, as detailed in Section 2.2 Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We used the official GEARS code base3 to reproduce the performance on the three benchmarking datasets. We need to rerun the experiments because the data splits used from the original paper were not published. Thus, directly comparing our results with the reported metrics from the GEARS paper is infeasible. Notes:	Comparison flag: RESOLVED — human adjudication selected Benchmark. Consensus status: RECONCILED Citation validity: Yes Method executed: Yes Uses (select all): Benchmark Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We used the official GEARS code base to reproduce the performance on the three benchmarking datasets. Resolution note: Human adjudication confirmed Benchmark as the primary use.
GEARS-HV-018 FORMAL VALIDATION xTrimoGene: An Efficient and Scalable Representation Learner for Single-Cell RNA-Seq Data DOI: 10.1101/2023.03.24.534055 Article page: https://doi.org/10.1101/2023.03.24.534055 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/03/25/2023.03.24.534055.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend/Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: While utilizing xTrimoGene, the normal state (before perturbation) gene expression profile is fed into xTrimoGene and we obtained the context embedding, which replaces raw expression value input in GEARS model (refer to Appendix 7.4 for detailed description).	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compared the native GEARS model with and without incorporating embeddings from xTrimoGene. Notes:	Comparison flag: RESOLVED — human adjudication selected Benchmark. Consensus status: RECONCILED Citation validity: Yes Method executed: Yes Uses (select all): Benchmark Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compared the native GEARS model with and without incorporating embeddings from xTrimoGene.

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	Notes:		Resolution note: Human adjudication confirmed Benchmark as the primary use.
GEARS-HV-019 FORMAL VALIDATION Knowledge-augmented Graph Machine Learning for Drug Discovery: A Survey DOI: 10.1145/3744237 Article page: https://doi.org/10.1145/3744237 Full text 1: https://pure.au.dk/ws/files/469052743/Knowledge-augmented_Graph_Machine_Learning_for_Drug_Discovery.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: GEARS [164] Nat. Biotechnol 2024 PredMol ✓ Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Mentioned in table Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-020 FORMAL VALIDATION Fluctuation structure predicts genome-wide perturbation outcomes DOI: 10.1101/2025.06.27.661814 Article page: https://doi.org/10.1101/2025.06.27.661814 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: At the same time, traditional linear regression models have some limitations: they typically treat genes as independent, cannot explicitly identify the genes driving the response (or the degree of impact per driver gene), and, with the exception of TRADE16, produce only point estimates of the regression coefficients and are not uncertainty-aware. Predictive models such as scGen20, GEARS21, and scGPT22, on the other hand, employ deep learning frameworks to model perturbation responses in latent	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We benchmarked against five recent approaches spanning distinct methodological classes—GenePert51, scLAMBDA21, Scouter52, scGPT20, and GEARS53—as well as a linear mean baseline, motivated by recent concerns that simple linear models can match far more complex architectures on this task54. T Notes:	Comparison flag: RECONCILE — differences: Method executed (No vs Yes) Consensus status: Citation validity: Yes Method executed: Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	spaces and can predict outcomes of unseen perturbations. Notes:		
GEARS-HV-021 FORMAL VALIDATION Multifaceted Representation of Genes via Deep Learning of Gene Expression Networks DOI: 10.1101/2024.03.07.583777 Article page: https://doi.org/10.1101/2024.03.07.583777 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/09/2024.03.07.583777.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Recent studies have employed advanced AI techniques to model the complexities of gene networks using large gene expression datasets1–11 Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Accurate predictive modeling of human gene relationships would fundamentally transform our ability to uncover the molecular mechanisms that underpin key biological and disease processes. Recent studies have employed advanced AI techniques to model the complexities of gene networks using large gene expression datasets1–11. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-022 FORMAL VALIDATION Learning Cross-Domain Representations for Transferable Drug Perturbations on Single-Cell Transcriptional Responses DOI: 10.1609/aaai.v39i18.34073 Article page: https://doi.org/10.1609/aaai.v39i18.34073 Full text 1: https://ojs.aaai.org/index.php/AAI/article/download/34073/36228	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Additionally, to evaluate our model's performance, we compared it against previously published methods, including chemCPA (Hetzel et al. 2022), GEARS (Roohani, Huang, and Leskovec 2022) and cycleCDR (Huang and Liu 2024). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Notably, these dual-gene perturbations were also chosen for testing in the GEARS study (Roohani, Huang, and Leskovec 2022), thus justifying a robust and objective performance comparison. We trained our model on the expr Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-023 FORMAL VALIDATION GPerturb: Gaussian process modelling of single-cell perturbation data DOI: 10.1038/s41467-025-61165-7 Article page: https://doi.org/10.1038/s41467-025-61165-7 Full text 1: https://www.nature.com/articles/s41467-025-61165-7.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We first compared the predictive performance of GPerturb, CPA, GEARS and SAMS-VAE on a subset of the genome-wide CRISPR interference Perturb-seq dataset. Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We first compared the predictive performance of GPerturb, CPA, GEARS and SAMS-VAE on a subset of the genome-wide CRISPR interference Perturb-seq dataset.33 Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-024 FORMAL VALIDATION scPRINT: pre-training on 50 million cells allows robust gene network predictions DOI: 10.1038/s41467-025-58699-1 Article page: https://doi.org/10.1038/s41467-025-58699-1 Full text 1: https://www.nature.com/articles/s41467-025-58699-1.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Understanding the cellular mechanism is considered a milestone in biology, allowing us to predict cell behavior and the impact of drugs and gene knock-outs1,2,3,4,5. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Understanding the cellular mechanism is considered a milestone in biology, allowing us to predict cell behavior and the impact of drugs and gene knock-outs1,2,3,4,5. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-025 FORMAL VALIDATION Mapping ovarian cancer spatial organization uncovers immune evasion drivers at the genetic, cellular, and tissue level DOI: 10.1101/2023.10.16.562592 Article page: https://doi.org/10.1101/2023.10.16.562592	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: As more Perturb-seq datasets, as the one generated here, become available across a more diverse range of cell types and conditions, it will be possible	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: it will be possible to use Perturb-seq data more effectively to extrapolate from one context to another with increasing accuracy70,71	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/19/2023.10.16.562592.full.pdf	to use Perturb-seq data more effectively to extrapolate from one context to another with increasing accuracy^{70,71} and, in the case of malignant cell states, using CNA-to-RNA and CNA-to-cell-state associations to further guide Perturb-seq experimental design. Notes:	Notes:	
GEARS-HV-026 FORMAL VALIDATION HyperCas12a-based multiplexed genetic regulation in <i>Candida albicans</i> DOI: 10.1093/nar/gkaf1402 Article page: https://doi.org/10.1093/nar/gkaf1402 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/10/20/2025.10.20.683421.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Recent technological advances in model systems have enabled the identification of a myriad of genetic interactions and non-additive phenotypic outcomes caused by changes in two or more genes [17–21]. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Recent technological advances in model systems have enabled the identification of a myriad of genetic interactions and non-additive phenotypic outcomes caused by changes in two or more genes [17–21]. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-027 FORMAL VALIDATION Prediction of cellular morphology change under perturbations with transcriptome-guided diffusion model DOI: 10.1101/2025.07.16.665106 Article page: https://doi.org/10.1101/2025.07.16.665106 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/16/2025.07.16.665106.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARS⁴⁷, is a possible extension. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARS⁴⁷, is a possible extension. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-028 FORMAL VALIDATION Causal models and prediction in cell line perturbation experiments DOI: 10.1186/s12859-024-06027-7 Article page: https://doi.org/10.1186/s12859-024-06027-7 Full text 1: https://www.researchsquare.com/article/rs-3973384/latest.pdf Full text 2: https://bmcbioinformatics.biomedcentral.com/counter/pdf/10.1186/s12859-024-06027-7	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: [10] developed GEARS to predict the effect of double knockouts/knockdowns using in vitro responses to single knockouts/knockdowns. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Several works have developed models for predicting responses to previously tested perturbations in new settings. For example, [7, 8], and [13] developed deep-learning autoencoder models to predict responses to previously untested perturbation-cell type combinations. [6] considered a similar problem and developed prediction model, SI-A, derived from the synthetic control literature in causal inference. [10] developed GEARS to predict the effect of double knockouts/knockdowns using in vitro responses to single knockouts/knockdowns. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-029 FORMAL VALIDATION HisToSpatialCNV: an interpretable deep learning method predicting spatial copy number variations from histopathology images DOI: 10.1038/s41551-026-01754-z Article page: https://doi.org/10.1038/s41551-026-01754-z Full text 1: https://www.nature.com/articles/s41551-026-01754-z.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Similar observations were also shared by others, in that single-cell FMs currently do not necessarily outperform established task-specific methods across all downstream tasks74,75,76,77. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Similar observations were also shared by others, in that single-cell FMs currently do not necessarily outperform established task-specific methods across all downstream tasks74,75,76,77. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-030 FORMAL VALIDATION Imputing spatial transcriptomics through gene network constructed from protein language model DOI: 10.1038/s42003-024-06964-2 Article page: https://doi.org/10.1038/s42003-024-06964-2 Full text 1: https://www.nature.com/articles/s42003-024-06964-2.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Most importantly, TransImpLR ignores gene-to-gene relationships in gene prediction, which have shown improved performance in previous work^{15,16}. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Most importantly, TransImpLR ignores gene-to-gene relationships in gene prediction, which have shown improved performance in previous work^{15,16}. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-031 FORMAL VALIDATION Simple Causal Relationships in Gene Expression Discovered through Deep Learned Collective Variables DOI: 10.1101/2023.01.18.524617 Article page: https://doi.org/10.1101/2023.01.18.524617 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/01/20/2023.01.18.524617.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: The use of autoencoders [9], UMAP [10,11], and other methods of dimensionality reduction [12,13] have demonstrated that it is possible to discover relationships in statistical covariation of many genes that can be used, for example, to distinguish different cell types and lineages [14], to predict mean gene expression under perturbation [15,16], to predict the interactions between genes under combinatorial perturbations [17], and to generate counterfactual predictions on single-cell response under new drug treatments [18] or novel perturbations [19]. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: The use of autoencoders [9], UMAP [10,11], and other methods of dimensionality reduction [12,13] have demonstrated that it is possible to discover relationships in statistical covariation of many genes that can be used, for example, to distinguish different cell types and lineages [14], to predict mean gene expression under perturbation [15,16], to predict the interactions between genes under combinatorial perturbations [17], and to generate counterfactual predictions on single-cell response under new drug treatments [18] or novel perturbations [19]. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-032 FORMAL VALIDATION Predicting transcriptional responses to novel chemical perturbations using deep generative model DOI: 10.21203/rs.3.rs-3917469/v1 Article page: https://doi.org/10.21203/rs.3.rs-3917469/v1 Full text 1: https://www.researchsquare.com/article/rs-3917469/latest.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: GEARS and CellOracle [11, 12] leveraged64 65 a knowledge graph of gene-gene relationships to predict genetic perturbation outcomes. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: r chemical perturbations of diverse cell 63 type and compound combinations. GEARS and CellOracle [11 , 12] leveraged 64 a knowledge graph of gene-gene relationships to predict genetic perturbation 65 outcomes. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-033 FORMAL VALIDATION Molecular causality in the advent of foundation models DOI: 10.1038/s44320-024-00041-w Article page: https://doi.org/10.1038/s44320-024-00041-w Full text 1: https://link.springer.com/content/pdf/10.1038/s44320-024-00041-w.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Hybrid models make a tradeoff between those extremes, which is why they have been found to be useful in systems biology, where data are currently scarce (AlQuraishi and Sorger, 2021;	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Hybrid models make a tradeoff between those extremes, which is why they have been found to be useful in systems biology, where data are currently scarce (AlQuraishi and Sorger, 2021;	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	Nilsson et al, 2022; Faure et al, 2023; Roohani et al, 2023; Fortelny and Bock, 2020; Lotfollahi et al, 2023; Yuan et al, 2021). Notes:	Nilsson et al, 2022; Faure et al, 2023; Roohani et al, 2023; Notes:	
GEARS-HV-034 FORMAL VALIDATION Knowledge graph-aided Bayesian active learning for top-K genetic interaction discovery DOI: 10.1038/s41598-025-13972-7 Article page: https://doi.org/10.1038/s41598-025-13972-7 Full text 1: https://www.nature.com/articles/s41598-025-13972-7.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For gene knockdown experiments, active learning strategies have been investigated in several studies3,4,5,6,7,8. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For gene knockdown experiments, active learning strategies have been investigated in several studies3,4,5,6,7,8. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-035 FORMAL VALIDATION Benchmarking foundation cell models for post-perturbation RNA-seq prediction DOI: 10.1186/s12864-025-11600-2 Article page: https://doi.org/10.1186/s12864-025-11600-2 Full text 1: https://bmcbgenomics.biomedcentral.com/counter/pdf/10.1186/s12864-025-11600-2	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We reproduced scGPT's results on the GEARS benchmarking datasets using the cell-gears v0.0.1 package and a fork of scGPT v0.2.1 (commit 7301b51). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We reproduced scGPT's results on the GEARS benchmarking datasets using the cell-gears v0.0.1 package and a fork of scGPT v0.2.1 (commit 7301b51). Notes:	Comparison flag: RESOLVED — human adjudication selected Mention only. Consensus status: RECONCILED Citation validity: Yes Method executed: No Uses (select all): Mention only Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We reproduced scGPT's results on the GEARS benchmarking datasets using the cell-gears v0.0.1 package and a fork of scGPT v0.2.1 (commit 7301b51). Resolution note: Human adjudication determined that the study used the GEARS datasets/package context but did not execute GEARS as the evaluated method.

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-036 FORMAL VALIDATION World Models for Pharmaceutical Science Should Simulate Biological Consequences rather than Merely Continue Statistical Patterns DOI: 10.51847/efvx6kqqna Article page: https://doi.org/10.51847/efvx6kqqna Full text 1: https://pharmacophorejournal.com/storage/files/article/5bd92b3d-cf40-41bc-80a2-5a1214db27a5-fP5jJbFj6Ln5v3zj/iZJfMPXRrTmfSTg.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Genetic interaction screens provide rich single-cell phenotypes for testing whether a model recovers nonlinear interaction structure; multiplex chemical transcriptomics measures heterogeneous cellular responses to pharmacological perturbations; and prediction of unseen multigene interventions tests whether a model generalizes beyond conditions explicitly represented during training [23-25]. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Genetic interaction screens provide rich single-cell phenotypes for testing whether a model recovers nonlinear interaction structure; multiplex chemical transcriptomics measures heterogeneous cellular responses to pharmacological perturbations; and prediction of unseen multigene interventions tests whether a model generalizes beyond conditions explicitly represented during training [23-25]. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-037 FORMAL VALIDATION Perturbomics: CRISPR–Cas screening-based functional genomics approach for drug target discovery DOI: 10.1038/s12276-025-01487-0 Article page: https://doi.org/10.1038/s12276-025-01487-0 Full text 1: https://www.nature.com/articles/s12276-025-01487-0.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Leskovec and colleagues, for instance, developed the graph-enhanced gene activation and repression simulator (GEARS), which integrates CRISPRa screening data with previously published Perturb-seq datasets to predict combinatorial perturbation effects on gene interactions100. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Leskovec and colleagues, for instance, developed the graph-enhanced gene activation and repression simulator (GEARS), which integrates CRISPRa screening data with previously published Perturb-seq datasets to predict combinatorial perturbation effects on gene interactions100. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
GEARS-HV-038 FORMAL VALIDATION scYeast: a Biological-knowledge-guided Foundation Model on Yeast Single-Cell Transcriptomics DOI: 10.1101/2025.08.20.671179 Article page: https://doi.org/10.1101/2025.08.20.671179 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/25/2025.08.20.671179.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Since scYeast has already integrated prior knowledge such as gene co-expression and transcriptional regulatory relationships during its pre-training, we adapted the GEARS [32] framework by using scYeast as the base model for predicting gene perturbation responses (Methods) (Fig. 4a). Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: we adapted the GEARS [32] framework by using scYeast as the base model for predicting gene perturbation responses (Methods) (Fig. 4a). Notes:	Comparison flag: MATCH — terminology normalization needed (“Extend” → “Extend or develop”). Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-039 FORMAL VALIDATION Cell Decoder: decoding cell identity with multi-scale explainable deep learning DOI: 10.1186/s13059-025-03832-y Article page: https://doi.org/10.1186/s13059-025-03832-y Full text 1: https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-025-03832-y	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Due to the incorporation of gene/protein interaction information, Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations [42]. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations [42]. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-040 FORMAL VALIDATION Medea: An AI agent for therapeutic reasoning across biological contexts DOI: 10.64898/2026.01.16.696667	Review status: LOCKED Citation validity: No Method executed: No Uses (select all): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator:	Review status: Citation validity: No Method executed: Uses (select all): Primary use: Biological insight: Evidence sufficiency: No Supporting quote + locator:	Comparison flag: RESOLVED — citation not located; exclude from the substantively classified dataset. Consensus status: RECONCILED Citation validity: Unclear Method executed: Unclear Uses (select all): Unclear Primary use: Unclear

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Article page: https://doi.org/10.64898/2026.01.16.696667 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2026/01/20/2026.01.16.696667.full.pdf	Notes:	Notes:	Biological insight: Unclear Evidence sufficiency: Citation not located Supporting quote + locator: Resolution note: Human adjudication: remove this record from the substantively classified dataset because the GEARS citation was not located.
GEARS-HV-041 FORMAL VALIDATION The manatee variational autoencoder model for predicting gene expression alterations caused by transcription factor perturbations DOI: 10.1038/s41598-024-62620-z Article page: https://doi.org/10.1038/s41598-024-62620-z Full text 1: https://www.nature.com/articles/s41598-024-62620-z.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: CPA3 autoencoder, together with GEARS4 and graphVCI5 graph neural network models disentangle perturbation covariates with basal cellular states, thus providing more flexibility for the in silico perturbation analysis. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: CPA3 autoencoder, together with GEARS4 and graphVCI5 graph neural network models disentangle perturbation covariates with basal cellular states, thus providing more flexibility for the in silico perturbation analysis. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-042 FORMAL VALIDATION Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT DOI: 10.1038/s42256-025-01170-z Article page: https://doi.org/10.1038/s42256-025-01170-z Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/04/23/2025.04.21.649754.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: It improved the mean squared error (MSE) and Pearson's correlation coefficient (PCC) for the top 20 most differentially expressed genes (MSE_DE and PCC_DE) by 0.06 and up to 0.15, respectively, compared with zero-shot scFoundation, and by 0.02–0.05 and	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For perturbation prediction using scFoundation and scGPT, scPEFT enabled scFoundation to overcome GPU memory limitations encountered during fine-tuning. It improved the mean squared error (MSE) and Pearson's correlation coefficient (PCC) for the top 20 most differentially	Comparison flag: RESOLVED — human adjudication confirmed that GEARS was executed. Consensus status: RECONCILED Citation validity: Yes Method executed: Yes Uses (select all): Benchmark Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: It improved the mean squared error (MSE) and Pearson's correlation coefficient (PCC) for the top 20 most differentially expressed genes by 0.02–0.05 and up

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	up to 0.04, respectively, compared with GEARS29 (Supplementary Table 5). Notes:	expressed genes (MSE_DE and PCC_DE) by 0.06 and up to 0.15, respectively, compared with zero-shot scFoundation, and by 0.02–0.05 and up to 0.04, respectively, compared with GEARS29 (Supplementary Table 5). These improvements highlight the importance of efficient adaptation for this task. scPEFT also outperformed a linear model baseline30 by 0.02 to 0.09 in MSE_DE and 0.02 to 0.15 in PCC_DE. However, none of the deep learning models surpassed the linear model when evaluated across all genes (Supplementary Note 9). scPEFT with the scGPT backbone performed comparably with fine-tuned scGPT and GEARS, except on the Norman dataset with two-gene perturbations (Supplementary Table 6). When evaluating all genes, no significant differences were observed among these models. Given its lower memory and time requirements, scPEFT offers a practical and accessible solution for predicting perturbation responses in the most differentially expressed genes. Notes:	to 0.04, respectively, compared with GEARS (Supplementary Table 5). Resolution note: Human adjudication confirmed execution; Benchmark remains the primary use.
GEARS-HV-043 FORMAL VALIDATION Predicting Multigene Transcriptional Outcomes Based on Dual-Stream Mamba Neural Network DOI: 10.12677/biphy.2025.131001 Article page: https://doi.org/10.12677/biphy.2025.131001 Full text 1: https://drpress.org/ojs/index.php/	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: The experimental results in Figure 4 demonstrate that the proposed model exhibits significant advantages across all data groups, with prediction	Review status: Citation validity: Yes Method executed: Uses (select all): Primary use: Biological insight: Evidence sufficiency: No Supporting quote + locator: Notes:	Comparison flag: RESOLVED — human adjudication followed Reviewer 1. Consensus status: RECONCILED Citation validity: Yes Method executed: Yes Uses (select all): Benchmark Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: The experimental results in Figure 4 demonstrate that the proposed model

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fcis/article/download/31264/30614 Full text 2: https://pdf.hanspub.org/biphy_2790105.pdf	performance substantially surpassing all baseline models Notes:		exhibits significant advantages across all data groups, with prediction performance substantially surpassing all baseline models. Resolution note: Human adjudication adopted Reviewer 1's classification and evidence-sufficiency assessment.
GEARS-HV-044 FORMAL VALIDATION BIOFORMERS: A SCALABLE FRAMEWORK FOR EXPLORING BIOSTATES USING TRANSFORMERS DOI: 10.1101/2023.11.29.569320 Article page: https://doi.org/10.1101/2023.11.29.569320 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/12/01/2023.11.29.569320.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: In the one gene perturbation experiments, we adhered to the preprocessing protocol established by Roohani et al. [26], which includes normalization by total molecular counts, log transformation of the data, selection of 5,000 highly variable molecules, and inclusion of any perturbed molecules otherwise omitted. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: . For this task, our benchmark dataset was the Perturb-seq dataset by Adamson et al. [25], which was preprocessed by Roohani et al. [26]. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-045 FORMAL VALIDATION Primer on machine learning applications in brain immunology DOI: 10.3389/fbinf.2025.1554010 Article page: https://doi.org/10.3389/fbinf.2025.1554010 Full text 1: https://www.frontiersin.org/journals/bioinformatics/articles/10.3389/fbinf.2025.1554010/pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For example, machine learning tools can be used to enhance dimensionality reduction and integration of large datasets, but also for more sophisticated tasks, such as predictive modeling of gene perturbations (Xu et	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For example, machine learning tools can be used to enhance dimensionality reduction and integration of large datasets, but also for more sophisticated tasks, such as predictive modeling of gene perturbations (Xu et	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
	al., 2021; Lotfollahi et al., 2022; Roohani et al., 2024). Notes:	al., 2021; Lotfollahi et al., 2022; Roohani et al., 2024). Notes:	
GEARS-HV-046 FORMAL VALIDATION Generative genomics accurately predicts future experimental results DOI: 10.1101/2025.09.08.674753 Article page: https://doi.org/10.1101/2025.09.08.674753 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/09/09/2025.09.08.674753.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Despite being a valuable resource for AI model development ^{20,34,35} , Perturb-seq data presents unique challenges, including technical artifacts and non-specific responses ^{20,36–39} , which require focused resolution for this data type that goes beyond the scope of our initial study. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Fourth, while our models encompass a wide array of gene expression experiments, we did not include extensive Perturb-seq data during training. Despite being a valuable resource for AI model development ^{20,34,35} , Perturb-seq data presents unique challenges, including technical artifacts and non-specific responses ^{20,36–39} , which require focused resolution for this data type that goes beyond the scope of our initial study. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-047 FORMAL VALIDATION Prediction of cellular morphology changes under perturbations with a transcriptome-guided diffusion model DOI: 10.1038/s41467-025-63478-z Article page: https://doi.org/10.1038/s41467-025-63478-z Full text 1: https://www.nature.com/articles/s41467-025-63478-z.pdf	Review status: Citation validity: Method executed: Uses (select all): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARS ⁴⁷ , is a possible extension. S Notes:	Comparison flag: RESOLVED — human adjudication selected Mention only. Consensus status: RECONCILED Citation validity: Yes Method executed: No Uses (select all): Mention only Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARS, is a possible extension.

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
			Resolution note: Human adjudication classified the record as Mention only.
GEARS-HV-048 FORMAL VALIDATION CellPLM: Pre-training of Cell Language Model Beyond Single Cells DOI: 10.1101/2023.10.03.560734 Article page: https://doi.org/10.1101/2023.10.03.560734 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.10.03.560734.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compare the performance between CellPLM and two baselines, i.e., a recent preprint GEARS method (Roohani et al., 2022), and scGen (Lotfollahi et al., 2019). Notes:	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare the performance between CellPLM and two baselines, i.e., a recent preprint GEARS method (Roohani et al., 2022), and scGen (Lotfollahi et al., 2019). Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-049 FORMAL VALIDATION Decoding cell identity with multi-scale explainable deep learning DOI: 10.1101/2024.02.05.578922 Article page: https://doi.org/10.1101/2024.02.05.578922 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/02/08/2024.02.05.578922.full.pdf	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations57. Notes:	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Due to the incorporation of gene/protein interaction information, Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations57. Notes:	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note:
GEARS-HV-050 FORMAL VALIDATION Perturbation Atlases as Pharmacological Reasoning Engines for Connecting Molecular Interventions to	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No	Comparison flag: MATCH — normalized core fields agree. Consensus status: Citation validity: Yes Method executed: No Uses (select all):

Article and approved evidence	Reviewer 1 independent record	Reviewer 2 independent record	Consensus human reference
Context-Dependent Cellular Responses DOI: 10.51847/a1xh4xuw5o Article page: https://doi.org/10.51847/a1xh4xuw5o Full text 1: https://pharmacophorejournal.com/storage/files/article/aa68d254-a799-4a9c-8a30-0f7cf119a88a-Y08w047JPZKDeQ4r/9rBBC2PqJpy90ji.pdf	Evidence sufficiency: Sufficient Supporting quote + locator: GEARS shows how graph-based information can contribute to prediction of transcriptional outcomes for novel multigene perturbations [16]. Notes:	Evidence sufficiency: Yes Supporting quote + locator: GEARS shows how graph-based information can contribute to prediction of transcriptional outcomes for novel multigene perturbations [16]. Notes:	Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Resolution note: